

Supplement 27.1 — $|\mathbb{N}| = |E|$

CS 173: Discrete Structures · Functions & Cardinality

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1 Setup

Definition 1.1 (Even Integers). Let E denote the set of **even integers**:

$$E = \{z \in \mathbb{Z} \mid (\exists k \in \mathbb{Z}) (2k = z)\} = \{\dots, -4, -2, 0, 2, 4, \dots\}.$$

Remark. We take $\mathbb{N} = \{0, 1, 2, \dots\}$. The goal is to prove $|\mathbb{N}| = |E|$ by constructing an explicit bijection $f: \mathbb{N} \rightarrow E$. The key idea is to *interleave*: map even-indexed naturals to the non-negative even integers $(0, 2, 4, \dots)$ and odd-indexed naturals to the negative even integers $(-2, -4, -6, \dots)$.

2 The Theorem

Theorem 2.1. $|\mathbb{N}| = |E|$.

Proof. Define $f: \mathbb{N} \rightarrow E$ by

$$f(n) = \begin{cases} n & \text{if } 2 \mid n, \\ -n - 1 & \text{if } 2 \nmid n. \end{cases}$$

Well-defined (maps into E).

- If $2 \mid n$, then $f(n) = n$ is even, so $f(n) \in E$.
- If $2 \nmid n$, then n is odd, so $n + 1$ is even, hence $f(n) = -(n + 1)$ is even, so $f(n) \in E$.

Injective. We must show $(\forall a, b \in \mathbb{N})(f(a) = f(b) \Rightarrow a = b)$. Let $a, b \in \mathbb{N}$ and assume $f(a) = f(b)$. We split on the sign of $f(a)$.

Case 1: $f(a) \geq 0$.

First we show $2 \mid a$ (i.e. a is even). Suppose, towards a contradiction, that $2 \nmid a$ (i.e. a is odd). Then by definition of f , $f(a) = -a - 1 \leq -1 < 0$ since $a \in \mathbb{N}$, contradicting $f(a) \geq 0$. Therefore $2 \mid a$.

Since $f(a) = f(b) \geq 0$, by a similar argument $2 \mid b$.

Since $2 \mid a$, by definition of f : $f(a) = a$. Since $2 \mid b$, by definition of f : $f(b) = b$. Therefore $a = f(a) = f(b) = b$, so $\boxed{a = b}$.

Case 2: $f(a) < 0$.

Suppose, towards a contradiction, that $2 \mid a$. Then $f(a) = a \geq 0$ since $a \in \mathbb{N}$, contradicting $f(a) < 0$. Therefore $2 \nmid a$, i.e. $2 \mid a - 1$.

Since $f(a) = f(b) < 0$, by a similar argument $2 \mid b - 1$.

By definition of f : $f(a) = -a - 1$ and $f(b) = -b - 1$. So $-a - 1 = -b - 1$, which gives $-a = -b$, so $\boxed{a = b}$.

In both cases $f(a) = f(b) \Rightarrow a = b$, so f is injective.

Surjective. We must show $(\forall y \in E)(\exists x \in \mathbb{N})(f(x) = y)$. Let $y \in E$. By definition of E , there exists $k \in \mathbb{Z}$ such that $2k = y$. We split on the sign of y .

Case 1: $y \geq 0$.

Then $2k = y \geq 0$ means $k \geq 0$, so $k \in \mathbb{N}$. Therefore $2k \in \mathbb{N}$. Take $x = 2k$. Since $2 \mid 2k$, by definition of f :

$$f(x) = f(2k) = 2k = y.$$

Case 2: $y < 0$.

Since $y \in \mathbb{Z}$ and $y < 0$, we have $y \leq -1$. Then $y + 1 \leq 0$, so $-(y + 1) \geq 0$. Since $-(y + 1) \in \mathbb{Z}$ and $-(y + 1) \geq 0$, we have $-(y + 1) \in \mathbb{N}$.

Now observe: since $y = 2k$,

$$-(y + 1) = -(2k + 1) = -2k - 1.$$

So

$$-(y + 1) - 1 = -2k - 1 - 1 = -2k - 2 = 2(-k - 1),$$

which shows $2 \mid -(y + 1) - 1$, i.e. $2 \nmid -(y + 1)$ (since $-(y + 1)$ is odd). Therefore by definition of f , with $x = -(y + 1)$:

$$f(x) = f(-(y + 1)) = -(-(y + 1)) - 1 = (y + 1) - 1 = y.$$

In both cases we found $x \in \mathbb{N}$ with $f(x) = y$, so f is surjective.

Therefore f is injective and f is surjective, so f is bijective by definition. Hence $|E| = |\mathbb{N}|$ by definition of cardinality equality. $\square$

3 Explicit Correspondence

Example. The first several values of f :

n	0	1	2	3	4	5	6	7	$\dots$
$f(n)$	0	-2	2	-4	4	-6	6	-8	$\dots$

Even naturals hit $0, 2, 4, \dots$ and odd naturals hit $-2, -4, -6, \dots$, together covering every element of E exactly once.